Bayesian Network Lab

Faisal Alburti | CS3081 Artificial Intelligence | Effat University

Calculations: likelihood and inference

Notation: R = Rain, M = Maintenance, T = Train, A = Appointment. A joint probability multiplies one factor per node:

P(R, M, T, A) = P(R) x P(M | R) x P(T | R, M) x P(A | T)

Step 3: normal day

No rain, no maintenance, train on time, attend the appointment.

P(none, no, on time, attend)
= 0.7 x 0.6 x 0.9 x 0.9
= **0.3402 = 34.02%**

Step 4: bad day

Heavy rain, maintenance yes, train delayed, miss the appointment.

P(heavy, yes, delayed, miss)
= 0.1 x 0.1 x 0.6 x 0.4
= **0.0024 = 0.24%**

Step 6: enumeration given a delayed train

Sum over the two Maintenance states for each Rain value. Appointment is hidden too, but summing attend and miss contributes a factor of 1.

Rain	P(rain, delayed) calculation	Weight
None	0.7 x (0.4 x 0.2 + 0.6 x 0.1)	0.098
Light	0.2 x (0.2 x 0.4 + 0.8 x 0.3)	0.064
Heavy	0.1 x (0.1 x 0.6 + 0.9 x 0.5)	0.051

P(delayed) = 0.098 + 0.064 + 0.051 = **0.213**
Normalization factor: alpha = 1 / 0.213 = 4.6948.

P(none | delayed) = 0.098 / 0.213 = **0.4601**
P(light | delayed) = 0.064 / 0.213 = **0.3005**
P(heavy | delayed) = 0.051 / 0.213 = **0.2394**
The normalized values sum to 1.0000.